

CHAPTER 27

External skull

The skull is the bony skeleton of the head. It houses the brain, the organs of special sense, and the upper parts of the respiratory and digestive systems, and provides attachments for many of the muscles of the head and neck. Movement is restricted and is found only in relation to the mandible at the temporomandibular joint and at the atlanto-occipital joint for movement of the head in relation to the neck.

The skull consists of the cranium, facial skeleton and mandible. The cranium may be subdivided into the calvaria (sometimes called the cranial vault) and the basicranium (cranial base). It encloses the brain, cranial nerves, meninges, blood vessels and cerebrospinal fluid within the cranial cavity. Internally, the cranial base can be divided into three regions corresponding to the floor of the anterior, middle and posterior cranial fossae. An alternative subdivision of the skull distinguishes between the neurocranium (calvaria and basicranium) and the viscerocranium (facial skeleton), which hangs down from the front of the neurocranium and houses the organs of sight, smell and taste, as well as the openings to the respiratory and digestive systems via the nose and mouth, respectively. In clinical parlance, the terms skull base and cranial base are frequently used as though they are synonymous. Indeed, inspection of the inferior surface of a skull confirms that the middle and posterior cranial bases are the same as the middle and posterior skull bases. However, the anterior cranial base and the anterior skull base should not be used interchangeably because the anterior skull base includes the inferior surface of the facial skeleton, whereas the internal surface of the anterior cranial base is the floor of the anterior cranial fossa and may be further subdivided into a midline (central) and two lateral parts. Developmentally, the parts of the skull where the bones are derived by endochondral ossification constitute the chondrocranium; the remaining bones are derived by intramembranous ossification.

The skull is the most complex bony structure in the body: the young adult skull is composed of an average of 28 separate bones, many of which are paired; some in the median plane are single, though symmetrical. Most of the vault bones are flat, and consist of two tables or plates of compact bone enclosing a narrow layer of relatively dense cancellous marrow (diploic bone). The marrow within the skull bones is a site of haemopoiesis, at least in the young individual. These bones form by intramembranous ossification of a highly vascular connective tissue membrane and have often been referred to as 'dermal' in deference to their alleged ancient phylogenetic origin. The inner table is thinner and more brittle while the outer tends to be thicker and more resilient; this is important to remember when examining fractures to the skull caused by either blunt or sharp trauma. This bone type is also of considerable value in cranial bone grafting in the young, where the two tables can be split to produce an autograft. The skull bones vary in thickness in different regions, with the bones of the viscerocranium being rather delicate and therefore particularly susceptible to fracture.

The majority of bones in the skull articulate via fibrous joints termed sutures, which facilitate growth rather than movement in the developing skull. The three main sutural morphologies reflect the magnitude of strain placed on them. Thus the margins of adjacent bones of a suture may be smooth and meet end to end, giving a simple (butt-end) suture (e.g. median palatine suture); may be bevelled, so that the border of one bone overlaps the other (e.g. parietotemporal suture); or may present numerous projections that interlock, giving a serrated jigsaw appearance (e.g. lambdoid suture). The complexity of serrated sutures increases from the inner to the outer surface. Fusion across sutures (synostosis) can start early in the third decade, although its variability precludes using this information to assess age with any degree of accuracy. The process of fusion starts on the internal surface of the cranium first and proceeds externally; the coronal and sagittal sutures are often the first to display synostosis. By middle age, many of the larger sutures will show evidence of synostosis, although there are some that rarely show fusion, e.g. the zygomaticofacial. Premature fusion of sutures

during the early growth phase of the skull will result in various abnormalities.

The bones forming the base of the skull develop mainly via endochondral ossification, and also play an important part in the overall growth of the face and the neurocranium. The joints between bones in the skull base are primary cartilaginous joints. One of the most important is the sphenoid-occipital synchondrosis, between the body of the sphenoid anteriorly and the basilar part of the occipital bone posteriorly; fusion between them is generally completed between 13 and 18 years of age.

There are only two sites of synovial articulation associated with the exterior of the skull, namely: the temporomandibular joint and the atlanto-occipital joint (between the condyles of the occipital bone and the superior articular facets of the atlas). Movement at these joints does not facilitate rotation but participates in flexion and extension in the anteroposterior plane and bilateral flexion in the transverse plane.

Many important neurological and vascular structures enter and exit the skull via foramina (openings). The skull is a prime site for fractures resulting from trauma, which means that these structures are at risk of damage following head injury. In addition to the main foramina, irregular emissary foramina allow veins situated externally on the face and scalp to communicate with those lying intracranially; spread of infection along these routes may have serious clinical consequences.

In the account of the skull that follows, only generalized standard views will be considered. A more detailed account of each individual bone will be found associated with the relevant regional text ([Berkovitz and Moxham 1994](#)).

FRONTAL (ANTERIOR) VIEW

Viewed from the front, the skull is generally ovoid in shape and is wider above than below ([Fig. 27.1](#)). The upper part is formed by the frontal bone, which underlies the forehead region above the orbits. Supero-medial to each orbit is a rounded superciliary arch (more pronounced in males), between which there may be a median elevation, the glabella. The glabella may show the remains of the interfrontal (metopic) suture, which usually closes in the first postnatal year ([Weinzweig et al 2003](#)), but persists in a small percentage of adult skulls in various ethnic groups. The frontal bone articulates with the two nasal bones at the frontonasal sutures, marked by a depression at the root of the nose; the point at which the frontonasal and internasal sutures meet is the anthropometric landmark known as the nasion.

The upper part of the face is occupied by the orbits and the bridge of the nose. Each orbital opening is roughly quadrangular in shape ([Ch. 41](#)). The upper, supraorbital, margin is formed entirely by the frontal bone, interrupted at the junction of its sharp lateral two-thirds and rounded medial third by the supraorbital notch or foramen, which transmits the supraorbital vessels and nerve. The lateral margin of the orbit is formed largely by the frontal process of the zygomatic bone and is completed above by the zygomatic process of the frontal bone; the suture between them (frontozygomatic) lies in a palpable depression. The infraorbital margin is formed by the zygomatic bone laterally and the maxilla medially. Both lateral and infraorbital margins are sharp and palpable. The medial margin of the orbit is formed above by the frontal bone and below by the lacrimal crest of the frontal process of the maxilla.

The central part of the face is occupied mainly by the paired maxillae, separated by the anterior nasal aperture. Each maxilla contributes to the upper jaw, the floor and medial wall of the orbital cavity, the lateral wall of the nose, the floor of the nasal aperture and the bone of the cheek. The anterior nasal spine marks the upper limit of the intermaxillary suture at the lower margin of the anterior nasal aperture and is palpable in the nasal septum. The infraorbital foramen transmits

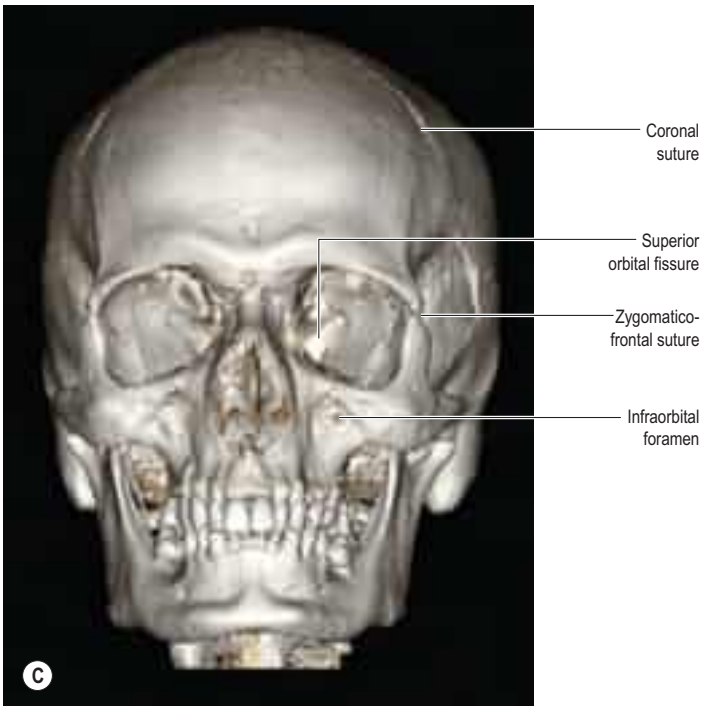
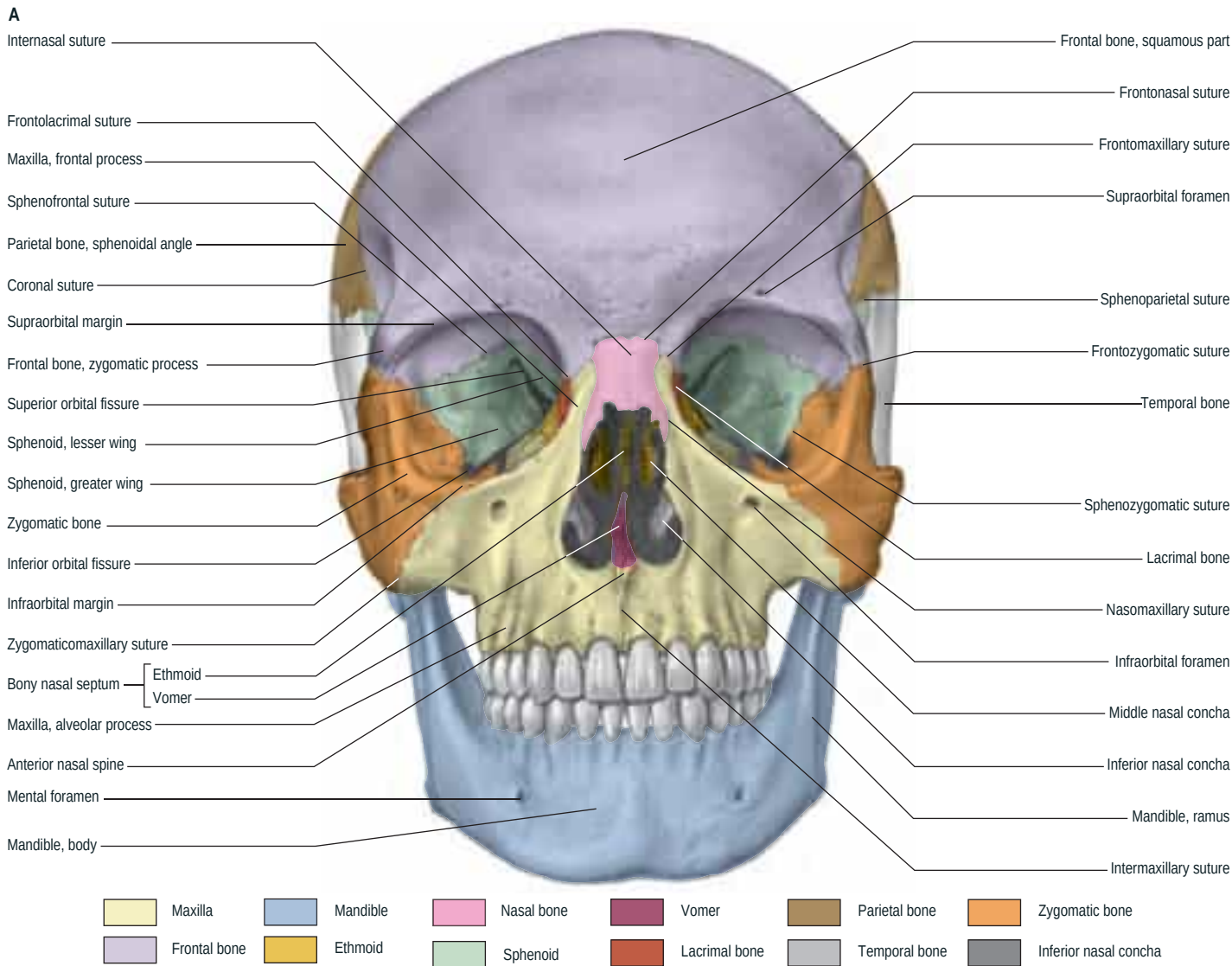


Fig. 27.1 **A**, The bones of the adult skull, frontal aspect. **B**, An anteroposterior view of the skull and upper cervical spine, young adult. Key: 1, sagittal suture; 2, frontal sinus; 3, crista galli; 4, mastoid air cells; 5, nasal septum; 6, inferior concha; 7, dens of axis. **C**, Reformatted volume rendered CT scan of anteroposterior view of the skull. (A, With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

the infraorbital vessels and nerve, and lies about 1 cm below the middle of the infraorbital margin. The maxillary alveolar process bears the upper teeth. The short, thick zygomatic process of the maxilla has an oblique upper surface that articulates with the zygomatic bone at the zygomaticomaxillary suture. The frontal process of the maxilla ascends posterolateral to the nasal bone to articulate with it medially, the frontal bone superiorly and the lacrimal bone laterally.

The anterior nasal aperture is piriform in shape, wider below than above and bounded by the paired nasal bones and maxillae. The upper boundary of the aperture is formed by the nasal bones while the remainder is formed by the maxillae. In life, several cartilages (septal, lateral nasal, major and minor alar) help to delineate two nasal cavities. The shape of these bones can be used quite successfully to predict the shape of the cartilaginous nose in forensic facial reconstructions (Wilkinson and Rynn 2012).

The lower part of the face, below the nose, is formed from the alveolar arch of the maxillae and the upper dentition, the body of the mandible, the alveolar process of the mandible and the lower dentition. In the midline, the mental protuberance produces the characteristic prominence of the chin. The mental foramen, which transmits the mental nerve and accompanying vessels, lies in the same vertical plane as the supraorbital and infraorbital foramina.

Anteroposterior radiographs of the skull clearly show the central location of the paranasal air sinuses in the frontal bone, maxilla and ethmoid. These can be particularly useful indicators of identity when postmortem images are compared with antemortem clinical films.

POSTERIOR VIEW

The parietal, temporal and occipital bones form the entirety of the posterior view (Fig. 27.2). The superolateral region is occupied by the parietal bones, the mastoid region of the temporal bones makes up the inferolateral regions, and the central portion is occupied by the occipital bone, which is the reason why this aspect is also referred to as the occipital view. The parietal bones articulate with the occipital bone at the lambdoid suture, which extends inferiorly into the occipitomastoid and the parietomastoid sutures behind and above the mastoid processes, respectively. The lambda is the anthropometric point where the occipital bone meets the two parietal bones. Accessory sutural bones are islands of bone that may be found within a suture. They may arise from separate centres of ossification and they appear to have no clinical significance, being of genetic rather than pathological aetiology; their initial formation is thought to be caused by a degree of dural strain and increased sutural width (Bellary et al 2013). A large, central interparietal bone is not uncommon and is sometimes referred to as an Inca bone (see Fig. 27.2).

The external occipital protuberance is a midline elevation on the occipital bone that can become particularly well developed and palpable in males. The location of the protuberance coincides with the anthropometric inion. Superior nuchal lines extend laterally from the protuberance and represent the boundary between the scalp and the neck. Inferior nuchal lines run parallel to, and below, the superior nuchal lines; a set of highest nuchal lines may sometimes occur above the superior lines. The external occipital protuberance, nuchal lines and roughened external surface of the occipital bone between the nuchal lines all afford attachment to muscles of the neck.

SUPERIOR VIEW

Seen from above, the contour of the calvaria varies greatly but is usually ellipsoid, or more strictly, a modified ovoid with its greatest width lying nearer to the occipital pole (Fig. 27.3). Four bones constitute this view and articulate via three well-defined sutures. The squamous part of the frontal bone is anterior, the squamous part of the occipital bone is posterior and the two parietal bones meet in the midline and separate the frontal from the occipital bone. The maximal parietal convexity on each site is palpable at the parietal tuber or eminence; it is most conspicuous in the female (retention of a pedomorphic appearance). The superior and inferior temporal lines run close to the parietal eminence but are best seen in a lateral view.

The coronal suture marks the articulation between the posterior margin of the frontal bone and the anterior margins of the two parietal bones. It descends across the calvaria and projects inferiorly until it meets the junction between the greater wing of the sphenoid and the squamous temporal bone at the pterion. The sagittal suture runs in the midline between the two parietal bones and extends from the bregma anteriorly to the lambda posteriorly. The lambdoid suture delineates the articulation between the posterior borders of the right and left parietal bones and the superior border of the occipital bone.

The bregma represents the position of the anterior fontanelle in the young child. This diamond-shaped, membrane-filled space located between the two frontal and two parietal bones of the developing fetal skull persists until approximately 18 months after birth. The lambda, at the junction of the sagittal and lambdoid sutures, represents the site of the posterior fontanelle, which persists for the first 2–3 months after birth.

A parietal foramen may pierce either or both parietal bones near the sagittal suture about 3.5 cm anterior to the lambda. It transmits a small emissary vein from the superior sagittal sinus. The vertex is the highest point on the skull and it usually occupies a position in the middle third of the sagittal suture.

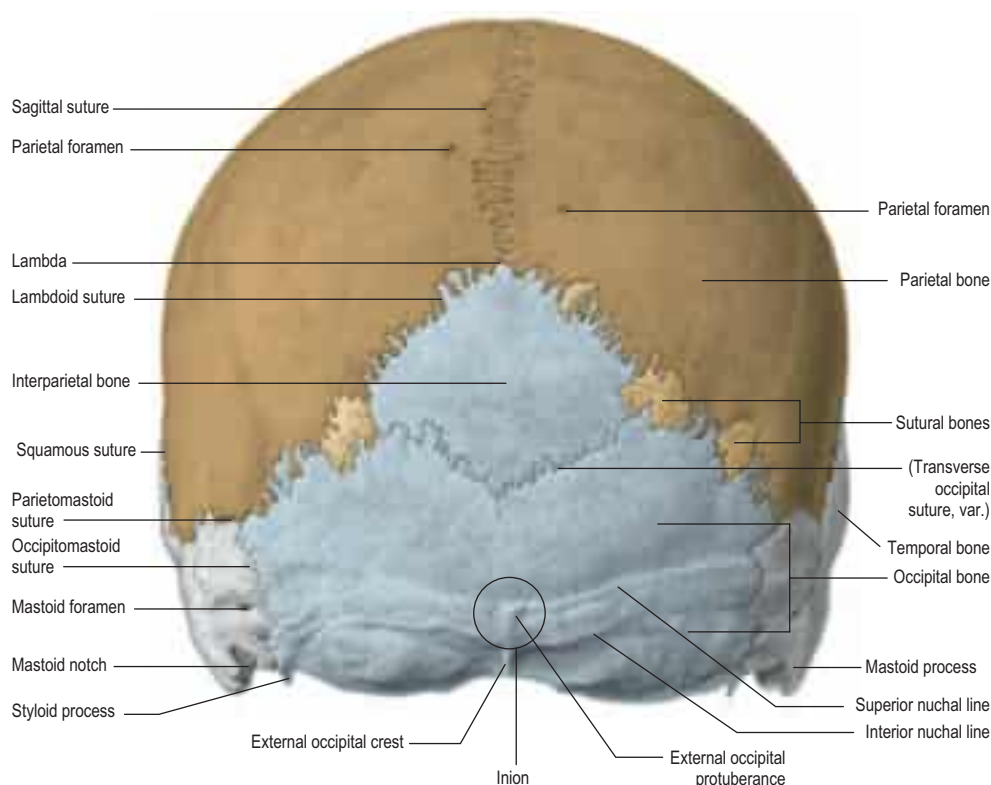


Fig. 27.2 The bones of the adult skull, posterior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)



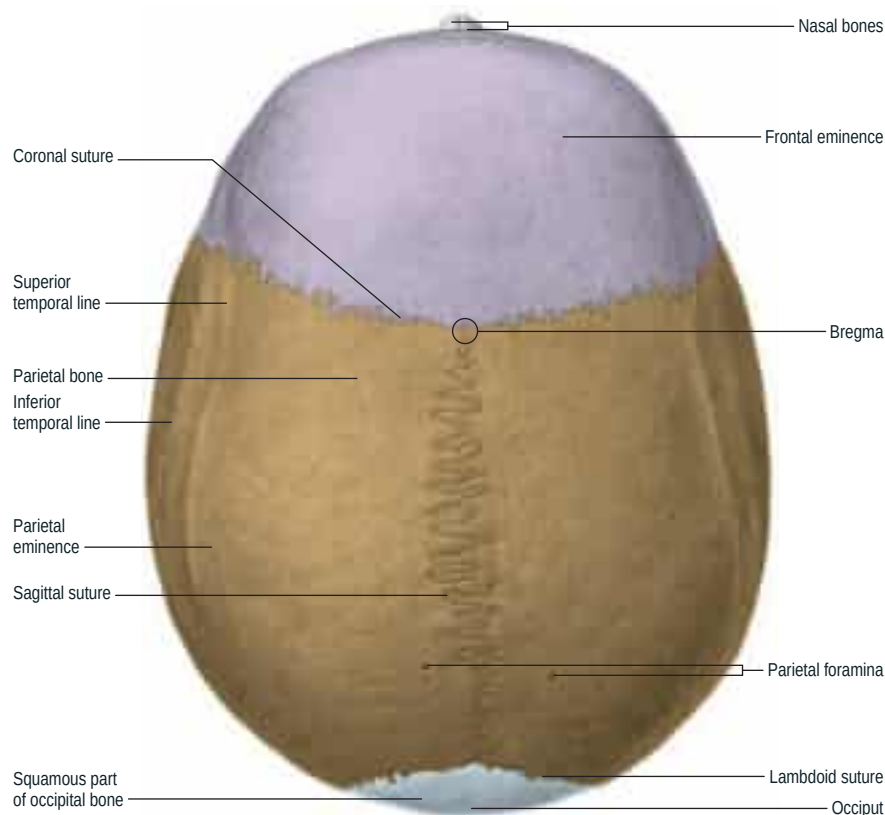


Fig. 27.3 The bones of the adult skull, superior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

LATERAL VIEW

The skull, viewed from the side, can be subdivided into three zones: face (anterior), temporal region (middle) and occipital region (posterior) (Fig. 27.4). The face has been considered in the section on the anterior view of the skull.

The temporal region can be divided into an upper temporal fossa and a lower infratemporal fossa, separated by the position of the zygomatic arch. The upper temporal fossa is bounded inferiorly by the zygomatic arch, superiorly and posteriorly by the temporal lines, and anteriorly by the frontal process of the zygomatic bone, and is continuous inferiorly with the infratemporal fossa deep to the zygomatic arch. The temporal lines often present anteriorly as distinct ridges but become much less prominent as they arch posteriorly across the parietal bone. The inferior temporal line becomes more prominent as it curves down the posterior part of the squamous temporal bone, forming a supra-mastoid crest at the base of the mastoid process. The superior temporal line gives attachment to the temporal fascia while the inferior temporal line provides attachment for temporalis.

The floor of the temporal fossa is formed by the frontal and parietal bones superiorly and the greater wing of the sphenoid and squamous part of the temporal bone inferiorly. All four bones of one side meet at a roughly H-shaped sutural junction termed the pterion. This is an important anthropometric landmark because it commonly overlies both the anterior branch of the middle meningeal artery and the lateral fissure of the cerebral hemisphere (Ma et al 2012). The pterion corresponds to the site of the anterolateral (sphenoidal) fontanelle of the neonatal skull, which closes in the third month after birth.

The vertical suture between the sphenoid and temporal bones, the sphenosquamosal suture, is formed by articulation between the posterior border of the greater wing of the sphenoid and the anterior border of the squamous part of the temporal bone.

The lateral surface of the ramus of the mandible will be described briefly here because it lies within the middle region of this view of the skull. The ramus is a plate of bone projecting upwards from the body of the mandible; its lateral surface gives attachment to masseter. The ramus bears two prominent processes superiorly, the coronoid process anteriorly and the condylar process posteriorly, separated by the mandibular notch. The coronoid process is the site of insertion of temporalis; the condylar process articulates with the mandibular fossa of the temporal bone at the temporomandibular joint. The inferior and posterior borders of the mandible meet at the angle; it is more commonly splayed in the male, reflecting the larger site of attachment for medial pterygoid on the internal surface.

The zygomatic arch stands proud of the rest of the skull, and the temporal and infratemporal fossae communicate via the gap thus created. In life, this space is largely filled by temporalis. The zygomatic bone is the principal bone of the cheek together with the zygomatic processes of the maxilla and temporal bones. The term 'zygomatic arch' is generally restricted to the temporal process of the zygomatic bone and the zygomatic process of the temporal bone, which articulate at the zygomaticotemporal suture. The suture between the zygomatic process of the frontal bone and the frontal process of the zygomatic bone is the frontozygomatic suture; the suture between the maxillary margin of the zygomatic bone and the zygomatic process of the maxilla is the zygomaticomaxillary suture; and the suture between the sphenoid and zygomatic bones is the sphenozygomatic suture. As the zygomatic process of the temporal bone passes posteriorly, it widens to form the articular tubercle of the mandibular fossa anteriorly.

The temporal bone is a prominent structure on the lateral aspect of the skull. Its squamous part lies in the floor of the upper temporal fossa and its zygomatic process contributes to the structure of the cheek. Additional components visible in the lateral view of the skull are the mandibular (glenoid) fossa and its articular eminence (tubercle), the tympanic plate, the external acoustic meatus (external auditory meatus), and the mastoid and styloid processes. The mandibular fossa is bounded in front by the articular eminence and behind by the tympanic plate. The articular eminence provides a surface over which the mandibular condyle glides during mandibular movements. The tympanic plate of the temporal bone contributes most of the margin of the external acoustic meatus; the squamous part forms the postero-superior region. The external margin is roughened to provide an attachment for the cartilaginous part of the meatus. A small depression, the suprameatal triangle, lies above and behind the meatus and is related to the lateral wall of the mastoid antrum. The mastoid process is an inferior projection of the temporal bone. It lies posteroinferior to the external acoustic meatus and is the site of attachment of sternocleidomastoid. It is in contact behind with the posteroinferior angle of the parietal bone at the parietomastoid suture and with the squamous part of the occipital bone at the occipitomastoid suture. These two sutures meet the lateral end of the lambdoid suture at the asterion. This coincides with the site of the posterolateral fontanelle in the neonatal skull, which closes during the second year. A mastoid foramen may be found near, or in, the occipitomastoid suture; it transmits an emissary vein from the sigmoid sinus. Sutural bones may appear in the parietomastoid suture.

The styloid process lies anterior and medial to the mastoid process and gives attachment to several muscles and ligaments. Its base is partly ensheathed by the tympanic plate and it descends anteromedially, its

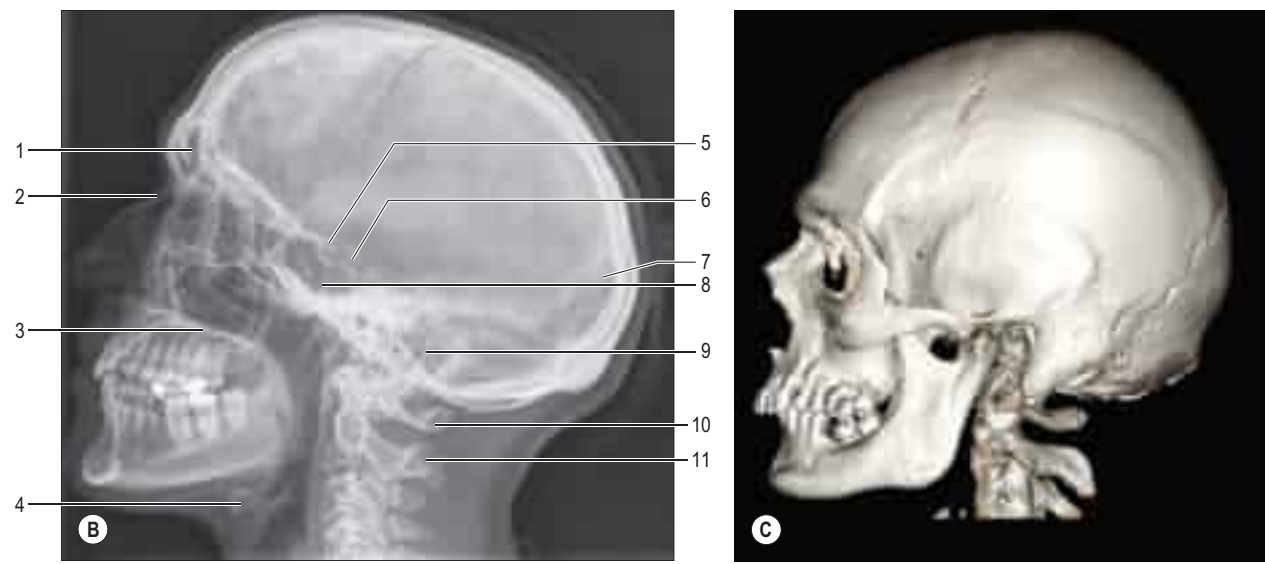
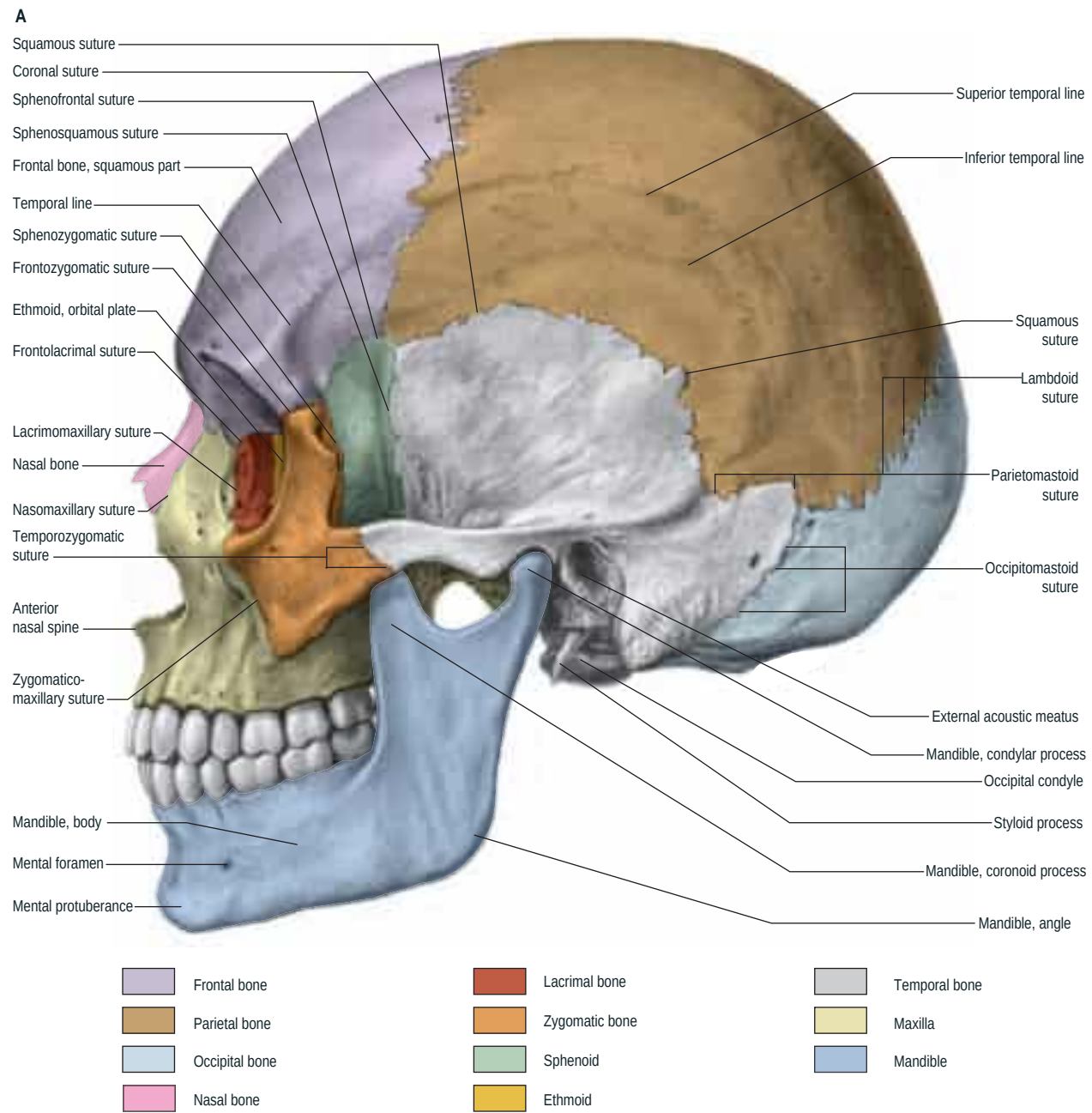


Fig. 27.4 A, The bones of the adult skull, lateral aspect. **B**, A true cephalometric lateral skull and cervical spine, young adult. **C**, Reformatted volume rendered CT scan of lateral view of the skull. (A, with permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

tip usually reaching a point medial to the posterior margin of the mandibular ramus. The styloid process is very variably developed. Its length ranges from a few millimetres to a few centimetres and increases with age (Krmpotić Nemančić et al 2009).

The infratemporal fossa is an irregular, postmaxillary space deep to the ramus of the mandible. It communicates with the upper temporal fossa deep to the zygomatic arch. It is best visualized when the mandible is removed but, for completeness, is considered here. Its roof is the infratemporal surface of the greater wing of the sphenoid, the lateral pterygoid plate lies medially, and the ramus of the mandible and styloid process lie laterally and posteriorly, respectively. The infratemporal fossa has no anatomical floor. Its anterior and medial walls are separated above by the pterygomaxillary fissure lying between the lateral pterygoid plate and the posterior wall of the maxilla. The infratemporal fossa communicates with the pterygopalatine fossa through the pterygomaxillary fissure.

INFERIOR VIEW

With the mandible removed, the inferior surface of the skull extends from the upper incisor teeth anteriorly to the superior nuchal lines of the occipital bone posteriorly (Fig. 27.5). The region contains many of the foramina through which structures enter and exit the cranial cavity.

The inferior surface may be conveniently subdivided into anterior, middle, posterior and lateral parts. The anterior part contains the hard palate and the dentition of the upper jaw, and lies at a lower level than the rest of the cranial base. The middle and posterior parts may be arbitrarily divided by a transverse plane passing through the anterior margin of the foramen magnum. The middle part is occupied mainly by the sphenoid bone, the apices of the petrous processes of the temporal bones, and the basilar part of the occipital bone. The lateral part contains the zygomatic arches, mandibular fossae, tympanic plates and the styloid and mastoid processes. The posterior part lies in the midline and is formed almost exclusively from the occipital bone. Whereas the middle and posterior parts are directly related to the cranial cavity (the middle and posterior cranial fossae), the anterior part is some distance from the anterior cranial fossa, being separated from it by the nasal cavities.

ANTERIOR PART OF INFERIOR SURFACE

The bony palate within the superior alveolar arch is formed by the palatine processes of the maxillae anteriorly and the horizontal plates of the palatine bones posteriorly, all meeting at a cruciform system of sutures (see Fig. 27.5). The median palatine suture runs anteroposteriorly and divides the palate into right and left halves. This suture is

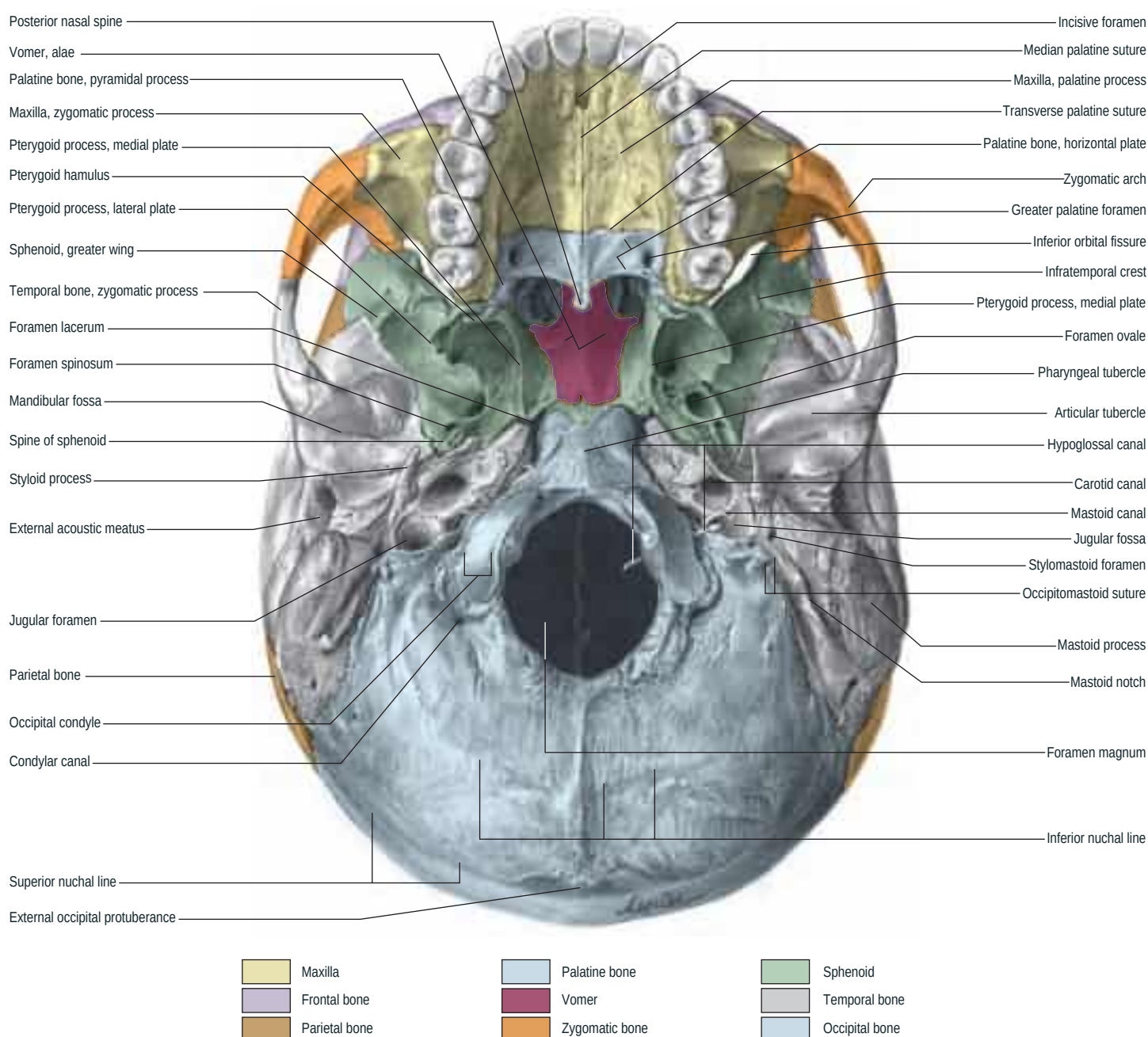


Fig. 27.5 The external surface of the base of the adult skull. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

continuous with the intermaxillary suture between the maxillary central incisor teeth. The transverse palatine (palatomaxillary) sutures run transversely across the palate between the maxillae and the palatine bones. The palate is arched sagittally and transversely; its depth and breadth are variable but are always greatest in the molar region. The incisive fossa lies behind the central incisor teeth, and the lateral incisive foramina, through which incisive canals pass to the nasal cavity, lie in its lateral walls. Median incisive foramina are present in some skulls and open on to the anterior and posterior walls of the fossa. The incisive fossa transmits the nasopalatine nerve and the termination of the greater palatine vessels. When median incisive foramina occur, the left nasopalatine nerve usually traverses the anterior foramen and the right nerve traverses the posterior foramen. The greater palatine foramen lies near the lateral palatal border of the transverse palatine suture; a vascular groove, deeper behind and shallower in front, leads forwards from the foramen. The lesser palatine foramina, usually two, lie behind the greater palatine foramen and pierce the pyramidal process of the palatine bone, which is wedged between the lower ends of the medial and lateral pterygoid plates. The palate is pierced by many other small foramina and is marked by pits for palatine glands. Variably prominent palatine crests extend medially from behind the greater palatine foramina. The posterior border projects back as the posterior nasal spine. In the adult, the alveolar arch normally bears a maximum of 16 sockets or alveoli for the teeth; the sockets vary in size and depth, and some are single and some are subdivided by septa, according to the morphology of the dental roots.

The nasal fossae lie above the hard palate and are separated by the nasal septum in the midline. The bony nasal septum is formed from the perpendicular plate of the ethmoid superiorly and the vomer inferiorly. The upper border of the vomer approximates to the inferior aspect of the body of the sphenoid, where it expands into an ala on each side. The two posterior nasal apertures (choanae) are located at the end of the nasal fossae; they are separated by the free posterior border of the vomer, and bounded below by the posterior border of the horizontal plates of the palatine bones, above by the sphenoid and laterally by the medial pterygoid plates.

MIDDLE PART OF INFERIOR SURFACE

The middle part of the inferior surface is made up from the sphenoid, petrous parts of the temporal bones and the basiocciput (see Fig. 27.5). It extends from the posterior nares anteriorly to an artificial line drawn transversely through the anterior margin of the foramen magnum posteriorly. In the adult, the body of the sphenoid fuses with the basiocciput to form a midline bar of bone that extends posteriorly to the foramen magnum (internally, this is known as the clivus). The basiocciput bears a small midline pharyngeal tubercle, which gives attachment to the pharyngeal raphe, the highest attachment of the superior pharyngeal constrictor.

The petrous processes of the two temporal bones pass from the lateral sides of the base of the skull and fill the triangular space between the greater wing of the sphenoid anteriorly and the lateral margins of the basiocciput posteriorly. Each petrous process meets the basilar part of the occipital bone at a petro-occipital junction, which is deficient posteriorly at the jugular foramen. The petrosphenoidal junction and the groove for the pharyngotympanic tube lie between the petrous process and the infratemporal surface of the greater wing of the sphenoid. The apex of the petrous process does not meet the sphenoccipital synchondrosis; the deficit so produced is the foramen lacerum.

Pterygoid processes descend from the junction between the greater wing and body of the sphenoid, separated by a pterygoid fossa. Anteriorly, the plates are fused, except inferiorly, where they are separated by the pyramidal process of the palatine bone. Sutures are usually discernible at this site in young skulls. Laterally, the pterygoid plates are separated from the posterior maxillary surface by the pterygomaxillary fissure, which leads into the pterygopalatine fossa. The posterior border of the medial pterygoid plate is sharp, and bears a small projection near the midpoint, above which it is curved and attached to the pharyngeal end of the pharyngotympanic tube. Above, the medial pterygoid plate divides to enclose the scaphoid fossa. Below, it projects as a slender pterygoid hamulus, which curves laterally and is grooved anteriorly by the tendon of tensor veli palatini. The pterygoid hamulus gives origin to the pterygomandibular raphe. The lateral pterygoid plate projects posterolaterally and its lateral surface forms the medial wall of the infratemporal fossa. Superiorly and laterally, the pterygoid process is continuous with the infratemporal surface of the greater wing of the sphenoid bone, which forms part of the roof of the infratemporal fossa. This surface forms the posterolateral border of the inferior orbital

fissure and bears an infratemporal crest associated with the origin of the upper part of lateral pterygoid. The infraorbital and zygomatic branches of the maxillary nerve and accompanying vessels pass through the inferior orbital fissure. Laterally, the greater wing of the sphenoid articulates with the squamous part of the temporal bone.

The medial aspect of the greater wing of the sphenoid presents a crescent of foramina of which only the most posterior two, foramen ovale and foramen spinosum, can be viewed on the basal aspect. The foramen ovale lies medial to the foramen spinosum and lateral to the foramen lacerum on the infratemporal surface of the greater wing of the sphenoid bone. It transmits the mandibular division of the trigeminal nerve, the lesser petrosal nerve, the accessory meningeal branch of the maxillary artery and an emissary vein that connects the cavernous venous sinus to the pterygoid venous plexus in the infratemporal fossa. Posterolaterally, the smaller and rounder foramen spinosum transmits the middle meningeal artery and a recurrent meningeal branch of the mandibular nerve. The irregular spine of the sphenoid projects posterolateral to the foramen spinosum. The medial surface of the spine is flat and, with the adjoining posterior border of the greater wing of the sphenoid, forms the anterolateral wall of a groove that is completed posteromedially by the petrous part of the temporal bone. This groove contains the cartilaginous pharyngotympanic tube, which leads posterolaterally into the bony portion of the tube that lies within the petrous part of the temporal bone. Occasionally, the foramen ovale and foramen spinosum are confluent or the posterior edge of the foramen spinosum may be defective. A small foramen, the sphenoidal emissary foramen (of Vesalius), is sometimes found between the foramen ovale and scaphoid fossa. When present, it contains an emissary vein linking the pterygoid venous plexus in the infratemporal fossa with the cavernous sinus in the middle cranial fossa.

The foramen lacerum is bounded in front by the body and adjoining roots of the pterygoid process and greater wing of the sphenoid bone; posterolaterally, by the apex of the petrous part of the temporal bone; and medially, by the basilar part of the occipital bone. Although nearly 1 cm long, it is not traversed by any major structure. The almost circular carotid canal lies behind and posterolateral to the foramen lacerum in the petrous part of the temporal bone. The internal carotid artery enters the skull through this foramen, ascends in the carotid canal, and turns anteromedially to reach the posterior wall of the foramen lacerum. It ascends through the upper end of the foramen lacerum, accompanied by venous and sympathetic nerve plexuses. Meningeal branches of the ascending pharyngeal artery and emissary veins from the cavernous sinus also traverse the foramen lacerum. In life, the lower part of the foramen lacerum is partially occluded by cartilaginous remnants of the embryological chondrocranium. The pterygoid canal can be seen on the base of the skull at the anterior margin of the foramen lacerum, above and between the pterygoid plates of the sphenoid bone. It leads into the pterygopalatine fossa and transmits the nerve of the pterygoid canal and accompanying blood vessels.

POSTERIOR PART OF INFERIOR SURFACE

The posterior part of the inferior surface of the skull is predominantly formed by the occipital bone (see Fig. 27.5). Prominent features are the foramen magnum and associated occipital condyles, jugular foramen, mastoid notch and the squamous part of the occipital bone up to the external occipital protuberance and the superior nuchal lines, hypoglossal canals (anterior condylar canals) and condylar canals (posterior condylar canals).

The foramen magnum lies in an anteromedian position and leads into the posterior cranial fossa. It is oval and wider behind, with its greatest diameter being anteroposterior. It contains the lower end of the medulla oblongata, meninges, cerebrospinal fluid, vertebral arteries and veins, and the accessory nerves; the apical ligament of the dens and the tectorial membrane pass through it to attach to the internal basiocciput. Anteriorly, the margin of the foramen magnum is slightly overlapped by the occipital condyles, which project down to articulate with the superior articular facets on the lateral masses of the atlas. Each occipital condyle is oval in outline and orientated obliquely so that its anterior end lies nearer the midline than its posterior end. It is markedly convex anteroposteriorly, less so transversely, and its medial aspect is roughened by ligamentous attachments. The hypoglossal canal, directed laterally and slightly forwards, traverses deep to each condyle and transmits the hypoglossal nerve, a meningeal branch of the ascending pharyngeal artery and an emissary vein from the basilar plexus. A depression, the condylar fossa, lies immediately posterior to the condyle and may contain a (posterior) condylar canal for an emissary vein from the sigmoid sinus. This fossa accommodates the posterior margin of the

atlas when the head is fully extended. A jugular process articulates with the petrous part of the temporal bone lateral to each condyle and its anterior free border forms the posterior boundary of the jugular foramen.

Laterally, the occipital bone approximates with the petrous part of the temporal bone anteriorly at the petro-occipital junction, and the mastoid process of the temporal bone more posteriorly at the occipito-mastoid suture. The jugular foramen, a large, irregular hiatus, lies at the posterior end of the petro-occipital junction between the jugular process of the occipital bone and the jugular fossa of the petrous part of the temporal bone (see Fig. 37.1C) (Vogl and Bisdas 2009). A number of important structures pass through this foramen: inferior petrosal sinus (anterior); glossopharyngeal, vagus and accessory cranial nerves (middle); and internal jugular vein (posterior). A mastoid canaliculus runs through the lateral wall of the jugular fossa and transmits the auricular branch of the vagus nerve. The canaliculus for the tympanic branch of the glossopharyngeal nerve lies on the ridge between the jugular fossa and the opening of the carotid canal. A small notch, related to the inferior glossopharyngeal ganglion, may be found medially, on the upper boundary of the jugular foramen (it is more easily identified internally). The orifice of the cochlear canaliculus may be found at the apex of the notch.

The squamous part of the occipital bone exhibits the external occipital protuberance, supreme, superior and inferior nuchal lines, and the external occipital crest, all of which lie in the midline, posterior to the foramen magnum. The region is roughened for the attachment of muscles whose primary function is extension of the skull at the neck.

LATERAL PART OF INFERIOR SURFACE

The lateral part of the inferior surface consists of the zygomatic arch and infratemporal fossa anteriorly and the mandibular fossa, tympanic plate and styloid and mastoid processes posteriorly (see Fig. 27.5). The anterior structures have been considered earlier in this chapter.

The mandibular fossa is a thin-walled, smooth, concave depression in the temporal bone and is most easily inspected when the mandible is removed. The zygomatic arch extends laterally in front of the fossa and a distinct ridge, the articular eminence, lies anterior to the fossa. Three fissures can be distinguished behind the mandibular fossa. The squamotympanic fissure extends from the spine of the sphenoid, between the mandibular fossa and the tympanic plate of the temporal bone, and curves up the anterior margin of the external acoustic meatus. A thin wedge of bone forming the inferior margin of the tegmen tympani lies within the fissure and divides the squamotympanic fissure into petrotympanic and petrosquamous fissures. The petrotympanic fissure transmits the chorda tympani branch of the facial nerve from the intracranial cavity into the infratemporal fossa. The tympanic plate forms the floor of the external acoustic meatus.

The stylomastoid foramen lies between the mastoid and styloid processes on the lateral aspect of the temporal bone. It transmits the facial nerve and the stylomastoid artery. The distance from the styloid process to the stylomastoid foramen decreases with age (Krmptović Nemančić et al 2009). A groove, the mastoid notch, lies medial to the mastoid process and gives origin to the posterior belly of digastric. A groove related to the occipital artery often lies medial to the mastoid notch. A mastoid foramen may be present near, or in, the occipitomastoid suture; when present, it transmits an emissary vein from the sigmoid sinus. The external acoustic meatus lies in front of the mastoid process. It is surrounded inferiorly by the tympanic plate, which partly ensheathes the base of the styloid process as the vaginal process.

CRANIAL FOSSAE (ANTERIOR, MIDDLE, POSTERIOR)

The cranial fossae are described in Chapter 28.

DISARTICULATED INDIVIDUAL BONES

Individual bones are described in the appropriate chapters. The bones of the facial skeleton and cranial vault are described in the chapters on the face and scalp (Ch. 30), nose and paranasal sinuses (Ch. 33), external and middle ear (Ch. 37), and orbit (Ch. 41). The sphenoid and mandible are described in the chapter on the infratemporal fossa (Ch. 32) and the occipital bone is described in the chapter on the back (Ch. 43).

JOINTS

The general characteristics of cranial sutures and the detailed anatomy of the temporomandibular joint and the atlanto-occipital joints are described in Chapter 5 and on page 541 and page 736, respectively. Sutural bones are described below on page 425.

NEONATAL, PAEDIATRIC AND SENESCENT ANATOMY

THE SKULL AT BIRTH

At birth, the calvaria is large in proportion to other skeletal parts, reflecting precocious cerebral maturation. The facial region is relatively small and constitutes only about one-eighth of the neonatal skull, whereas it constitutes about half of the adult skull (Fig. 27.6). Smallness of the face at birth is due mainly to the rudimentary stage of development of the mandible and maxillae; the teeth are unerupted and the maxillae are not yet pneumatized (invaded by air cells). The nose lies almost entirely between the orbits, and the lower border of the nasal aperture is only slightly lower in position than the orbital floors. Bones of the calvaria are unilaminar and lack diploë. Frontal and parietal tuberosities are prominent; in the frontal view, the greatest width occurs between the parietal tuberosities. The glabella, superciliary arches and mastoid processes are not developed and the cranial base is relatively short and narrow.

Ossification is incomplete, and many bones are still in several elements that are united by fibrous tissue or cartilage. The 'os incisivum' is continuous with the maxilla; pre- and postsphenoids may have just united, but the two halves of the frontal bone and the mandible, and the squamous, lateral and basilar parts of the occipital bone, are all separate, as are the petromastoid and squamotympanic parts of the temporal bone. The fibrous membrane that forms the calvaria remains unossified at the six angles of the parietal bones, producing six fontanelles: two single midline (anterior and posterior) and two lateral pairs (sphenoidal/anterolateral and mastoid/posterolateral). The anterior fontanelle is the largest, measuring approximately 4 cm in anteroposterior and 2.5 cm in transverse dimensions. It occupies the junction between the sagittal, coronal and frontal sutures and is rhomboid in shape. The posterior fontanelle lies at the junction between the sagittal and lambdoid sutures and is triangular in shape. The sphenoidal (anterolateral) and mastoid (posterolateral) fontanelles are small and irregular in shape and lie at the sphenoidal and mastoid angles of the parietal bones, respectively.

At birth, the orbits appear relatively large. The developing tooth germs are generally contained within the alveolar crypts, although eruption of the upper central incisor teeth can occur prior to, or shortly after, birth. Temporal bones differ greatly from their adult form. The internal ear, tympanic cavity, auditory ossicles and mastoid antrum are all almost adult in size, the tympanic plate is an incomplete ring that has usually started to fuse with the squamous part, and the mastoid process is absent. The external acoustic meatus is short, straight and wholly cartilaginous. The external aspect of the tympanic membrane faces more inferiorly than laterally, in accord with the basal cranial contour. The stylomastoid foramen is exposed on the lateral surface of the skull, the styloid process has not yet commenced ossification, the mandibular fossa is flat and more lateral, and its articular tubercle is undeveloped. The paranasal sinuses are rudimentary or absent and only the maxillary sinuses are usually identifiable.

The skull is moulded by slow compression during vaginal birth. That part of the scalp that is more central in the birth canal is often temporarily oedematous (caput succedaneum) as a result of interference with venous return. Fontanelles and the openness and width of the sutures allow bones of the cranial vault some overlap during the process of birth; the skull is compressed in one plane with compensatory orthogonal elongation. These deformations disappear within the first week after birth.

POSTNATAL GROWTH

Although postnatal growth of the calvaria (cranial vault) and facial skeleton usually proceeds in a coordinated fashion, these regions exhibit different rates and temporal periods of expression. In general, early growth of the vault is largely influenced by cerebral maturation while growth of the facial skeleton responds to spatial requirement for the development of the teeth and the muscles of mastication. Growth



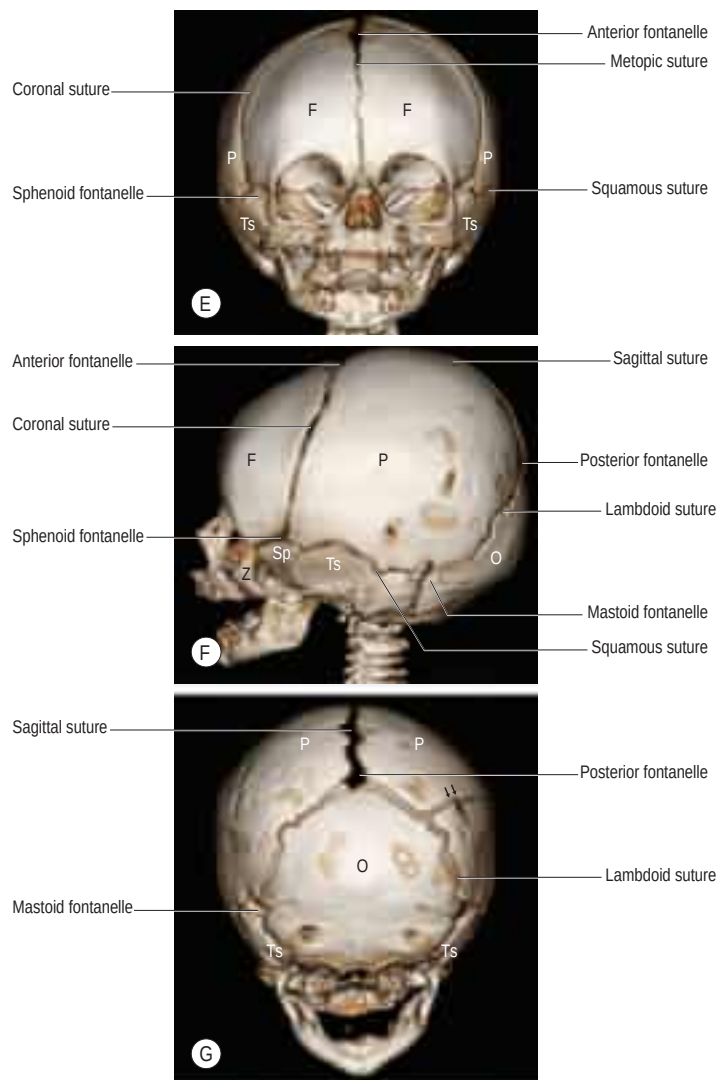


Fig. 27.6, E–G, Normal three-dimensional calvarial anatomy. E, Frontal aspect. F, Lateral aspect. G, Basal aspect; there is a linear fracture in the right parietal bone (double arrows). Abbreviations: F, frontal bone; O, occipital bone; P, parietal bone; Sp, sphenoid bone; Ts, temporal bone, squamous portion. (E–G, With permission from Y. Pekcevik, E. Hasbay, R. Pekcevik; Izmir/TR; Diagnostic value of three-dimensional CT in pediatric calvarial pathologies, ECR 2013 / C-1263, EPOS™, European Society of Radiology.)

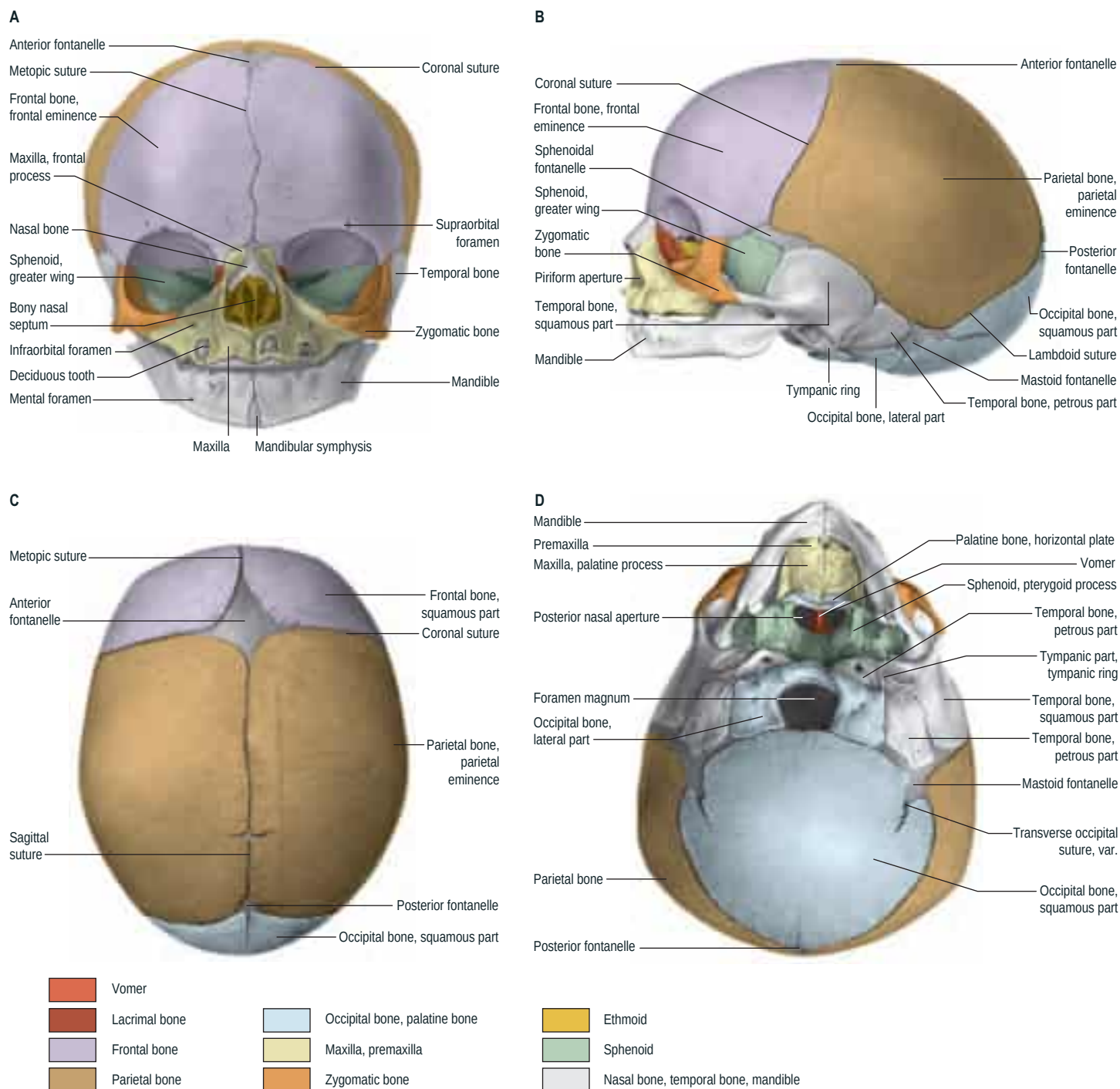


Fig. 27.6 A–D, The neonatal skull. **A**, Frontal aspect. **B**, Lateral aspect. **C**, Superior aspect. **D**, Basal aspect. (A–D, With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

of the cranial base is largely independent of both vault and facial development, and therefore the three regions will be considered separately. (For further reading, see [Belden \(1998\)](#), [Sgouros et al \(1999\)](#), [Scheuer and Black \(2000\)](#) and [Sperber and Guttmann \(2001\)](#).)

Growth of the calvaria

Growth of the calvaria is rapid during the first year and then continues at a slower rate until the seventh year, when it has reached almost adult dimensions. For most of this period, expansion is largely concentric; overall form is determined early in the first year and remains largely unaltered thereafter. However, the shape of the vault is not solely related to cerebral growth, but is also influenced by genetic factors that manifest in an extensive range of shapes and sizes that may be sufficient to allow an evaluation of ethnic origin. During the first and early second years, growth occurs primarily through ossification at apposed margins of bones (which possess an osteogenic layer), accompanied by some accretion and absorption of bone at surfaces in order to adapt to continually altering curvatures. Growth in breadth is said to occur at the

sagittal, sphenofrontal, sphenotemporal, occipitomastoid and petro-occipital junctions, while growth in height is said to occur at the fronto-zygomatic and squamosal sutures, pterion and asterion. During this period, fontanelles are closed by progressive ossification of the bones around them, but separate isolated centres of ossification may develop into sutural bones. The sphenoidal and posterior fontanelles close within 2 or 3 months of birth and the mastoid fontanelles usually close near the end of the first year. Wide variability in both the size and the timing of closure of the anterior fontanelle has been reported, but most will have closed by the middle of the second year and almost all will have closed by 24 months ([Pindrik et al 2014](#)).

Early in the first years, the calvarial bones commence interlocking at sutural junctions. Further expansion is largely achieved by accretion and absorption on external and internal bone surfaces, respectively. At the same time the bones also thicken, although this is not a uniform process. At birth, the vault is unilaminar but the tabular structure with intervening diploë is generally apparent soon thereafter. Thickening of the vault and development of external muscular markings reflect the influences of musculoskeletal maturation. The mastoid processes do not usually develop until the second year and they are pneumatized in

the sixth year (Scheuer and Black 2000). (For further reading about craniofacial morphology in preterm infants, see Paulsson and Bondemark 2009.)

Growth of the cranial base

Growth of the cranial base is largely responsible for cranial lengthening. It occurs at the cartilaginous joints between the sphenoid and ethmoid anteriorly, and especially between the sphenoid and occipital bones posteriorly (Lieberman et al 2000). Most of the growth occurs during the first five years of life (Sgouros et al 1999); it continues at the spheno-occipital synchondrosis until mid-teenage years. The prolonged period of continued expansion reflects a significant delaying mechanism that allows the continued and relatively late eruption of the permanent dentition. There is some evidence that growth may cease at about 15 years following eruption of the second molars, and this can often lead to insufficient space for growth of the third molars. A pubertal growth spurt has been ascribed to both sexes; it tends to occur some 2 years earlier in females and is likely to continue until the eighteenth year in the male. The anterior cranial base is significantly shorter in extremely preterm infants (under 28 weeks) (Paulsson and Bondemark 2009).

Growth of the facial skeleton

Growth of the facial skeleton occurs over a longer time period than is witnessed for the calvaria. The ethmoid and the orbital and upper nasal cavities have almost completed growth by the seventh year. Orbital and upper nasal growth is achieved by sutural accretion, with deposition of bone preferentially occurring on the facial aspects of the sutural junctions. The maxilla is carried downwards and forwards by expansion of the orbits and nasal septum and by sutural growth, especially at the fontanelles and zygomaticomaxillary and pterygomaxillary sutures. In the first year, growth in width occurs at the symphysis menti and mid-palatal, internasal and frontal sutures; such growth diminishes or even ceases when the symphysis menti and frontal suture close during the first few years, even though the mid-palatal suture persists until mature years. Facial growth continues up to puberty and shows a period of expansion that is linked to the growth spurt and hormonal influences of secondary sexual alteration. After sutural growth, near the end of the second year, expansion of the facial skeleton occurs by surface accretion on the face, alveolar processes and palate, and resorption in the walls of the maxillary sinuses, the upper surface of the hard palate and the labial aspect of the alveolar process. Coordinated growth and divergence of the pterygoid processes reflect deposition and resorption of bone on appropriate surfaces.

Obliteration of the calvarial sutures progresses with age, starting between 20 and 30 years internally, and somewhat later on the exterior. Closure times vary greatly. Obliteration usually begins in the coronal or sagittal sutures and then extends into the lambdoid suture. The size of the mandible and maxillae diminish following the loss of teeth and consequent resorption of alveolar bone; this reduces the vertical depth of the face and increases the mandibular angles.

Sutural bones

Additional ossification centres may occur in or near sutures, giving rise to isolated sutural bones (also called Wormian bones) (Bellary et al 2013). Usually irregular in size and shape, and most frequent in the lambdoid suture, they also occur at fontanelles, especially the posterior fontanelle. They may represent a pre-interparietal element, a true interparietal, or a composite. An isolated bone at the lambda is sometimes referred to as an Inca bone or Goethe's ossicle. One or more pterion ossicles or epipteric bones may appear between the sphenoidal angle of the parietal and the greater wing of the sphenoid; they vary greatly in size but are more or less symmetrical. Sutural bones usually have little morphological significance. However, they appear in great numbers in hydrocephalic skulls (Fig. 27.7), and they have therefore been linked with rapid cranial expansion.

Craniosynostosis

Sutural growth makes an important contribution to growth of the skull, especially during the first few years of life. Premature closure of sutures (craniosynostosis) leads to restriction of growth along the sutures, producing morphological changes that may result in skull deformity (Sharma 2013). Biomechanical forces and genetically determined local



Fig. 27.7 The hydrocephalic skull of a 25-year-old male showing numerous sutural bones. **A**, Lateral view. **B**, Posterior view. (Courtesy of the Museum of the Royal College of Surgeons of England, photograph by Mr J Carr.)

expression of growth factors have been implicated in the aetiology of premature fusion but the reasons are not fully understood. Premature fusion may occur in one or more of the cranial sutures; when the sutures around the cranial base are involved, severe limitation of facial bone growth will occur. Metabolic disorders such as rickets and familial hypophosphatasia can also result in synostosis. Raised intracranial pressure with or without hydrocephalus, visual deterioration and mental retardation may result. Scaphocephaly (sagittal craniosynostosis) is the most common and leads to lengthening of the vault in an anteroposterior direction; it can also occur in conjunction with other sutures, e.g. Crouzon's syndrome. Coronal synostosis, either unilateral (plagiocephaly) or bilateral (brachycephaly/oxycephaly), is the next most frequently seen and results in reduced anteroposterior development with marked supraorbital recession. When it is unilateral, the face develops asymmetrically and is rotated away from the side with premature fusion. Metopic craniosynostosis (trigonocephaly) and pansynostosis (turri-cephaly), where both the coronal and sphenofrontal sutures are involved, are much less common.

Craniofacial dysostosis syndromes, such as Crouzon's, Apert's, Saethre-Chotzen, Pfeiffer's and Carpenter's, show varying degrees of calvarial synostoses that are usually accompanied by a significant lack of growth in the midface. Early release of the calvarial synostoses does not result in normal facial growth, and a midfacial osteotomy at the Le Fort III level is usually required later in life. When significant orbital hypertelorism develops, a transcranial bipartitioning procedure is needed in order to bring the two orbits together.

Altered patterns of skull growth resulting in altered profiles can also be achieved deliberately by affecting sutural growth using binding and other pressure, as has been practised in certain cultures of the world (Fig. 27.8) (Tubbs et al 2006). The earliest known record of the practice is from Iraq and dates back to 45,000 BCE (Gerszten and Gerszten 1995).



Fig. 27.8 Intentional skull modification: characteristic elongation of the cranial vault produced by head-binding of a newborn. (Courtesy of John Langdon.)

Congenital abnormalities affecting the skull

A large number of malformations and anomalies affect the bones and associated soft tissue structures of the skull and are the result of a localized error of morphogenesis during embryological development. Many are recognized patterns of malformation that are presumed to have the same aetiology. They do not arise as the result of just one isolated error in morphogenesis and are described as syndromes; specific texts should be consulted for further details.

IDENTIFICATION FROM THE SKULL

There are many ways of identifying an individual; in physical and forensic anthropology, the most important concern biological and personal identity (Wilkinson and Rynn 2012). Biological identity pertains to those features that allow an individual to be classified in relation to features present in other individuals, e.g. sex, age, ethnic origin and stature, whereas personal identity establishes criteria that are characteristic and discriminatory for a particular individual, e.g. DNA, fingerprints and dental information. The skull is a useful source of information for the establishment of both biological and personal identity and is probably the most studied region of the skeleton. The foundation of this obsession has many historical roots, but fundamentally it has arisen from the importance that humans place on the concept that the skull is the repository of 'self', and that it is the means by which interpersonal communication is effected. Our face is our primary means of recognition and communication, and therefore it plays a pivotal role in establishing and reconstructing the identity of an individual (Lahr 1996).

SEX DETERMINATION

The determination of sex from a juvenile skull is notoriously unreliable. While sexual differences have been detected in measurements of the mandible, orbits, tooth size and pattern of dental eruption, they do not reach a level of statistical discrimination that will allow accurate and reliable assessment. It is equally difficult to assign sex to the face of a child because the faces of prepubertal boys and girls are comparable; perceivable sexual dimorphism is not generally manifest until secondary sexual changes are completed. Growth in the female face ceases in advance of that of the male, and consequently, female sex-related characteristics are more pedomorphic. The defining characteristics of sex in an adult skull are therefore male in orientation and reflect the effects of the increased mass of the muscles of mastication, which attach to the mandible, and the muscles associated with maintaining the erect head. It is reported that using the skull alone, sex can be predicted with

over 80% accuracy in the adult. This is extremely encouraging, given that research has shown that the correct sex can usually be predicted from the adult living face with only 96% accuracy.

Generally speaking, the male skull is more robust and the female more gracile, although there are obvious genetic, and therefore ethnic, variations that must be considered when attempting to assign sex from a skull. The female forehead is generally higher, more vertical and more rounded than that of the male, with retention of the frontal eminences. The male mandible is more robust and larger than that of the female; it generally displays a greater height in the region of the symphysis menti, the chin is squarer, the condyles are larger, the muscle attachments are more pronounced and the gonial angle is generally less than 125°. A male skull has thicker and more rounded orbital margins, pronounced supraorbital ridges, and often a well-defined glabella that occupies the midline above the root of the nose. The temporal lines are more pronounced in the male and the supramastoid crest generally extends posterior to the external acoustic meatus.

Other sites of muscle attachment on the skull reflect the biomechanical requirement to keep the more robust male head erect. They include the mastoid process for sternocleidomastoid, which is generally more robust in the male and more gracile in the female, and the nuchal lines, especially the external occipital protuberance for the attachment of the ligamentum nuchae. The cranial base in the male is generally more robust and the bone is thicker, which means that this area of the skull survives inhumation particularly well and is therefore of value in sex identification from fragmentary remains.

AGE DETERMINATION

Age is a continuous variable; to establish chronological age from a skull requires that structures change with age at a relatively constant and predictable rate. The relationship between chronological age and skeletal maturity is closest in the juvenile years, and therefore greater accuracy is achieved in the prediction of age from the juvenile than from the adult skull.

The neonatal skull has been described above. Examples of additional features that allow reliable age determination throughout the subadult years are: development of the nasal spine (by year 3), completion of the hypoglossal canal (by year 4), formation of the foramen of Huschke (by year 5), ossification of the dorsum sellae (by year 5), and fusion of the different parts of the occipital bone (by year 7). The fontanelles are usually all closed by the middle of the second year; the posterolateral is the first to close in the first 2 months after birth, and the anterior fontanelle is the last to close around the middle of the second year. The mastoid process appears in the second year and the metopic suture between the two frontal bones is usually closed by the end of the first year. The sphenoccipital synchondrosis will fuse between 11 and 16 years in the female and 13 and 18 years in the male, while the vomer and the ethmoid will fuse between 20 and 30 years of age. The last part of the skull to show active age-related growth is the jugular growth plate, a small triangular area sited posterolateral to the jugular foramen in the occipitotemporal suture. Fusion here does not begin until the third decade, and bilateral fusion may not be completed before the middle of the fourth decade; in a small proportion of individuals, the plate may remain unfused into the sixth decade. Closure of the cranial sutures is age-related but the correlation is not strong and displays strong genetic variation. Thus, while it can be said that suture closure may begin in the early part of the third decade, and it is likely that many of the sutures will be obliterated with advancing age, it is not a reliable means of establishing the age of an individual. See Scheuer and Black (2000) for further information on age determination from the juvenile skeleton.

The most accurate means of determining age from the skull (of both a living and a deceased individual) is by assessment of dental maturation. Tooth development can be studied throughout the entire juvenile age span (from the early embryo to the adolescent), and importantly, dental age and chronological age have been shown to exhibit a stronger correlation than skeletal and chronological age. Further, the teeth tend to survive inhumation successfully and are remarkably resilient to fire and explosion, ensuring their value in forensic investigations.

The chronological pattern of dental maturation is well documented and is an extremely important tool for age evaluation. Tooth development can be separated into a number of well-defined stages: deciduous mineralization (crown and root), deciduous emergence and maturation, deciduous root resorption, shedding of deciduous teeth, mineralization (crown and root) of permanent dentition, emergence and maturation of deciduous dentition, and attrition of permanent crowns. These stages do not occur in a linear fashion; while some of the

deciduous teeth are emerging, permanent teeth are already being formed. For example, mineralization of the deciduous central incisor commences around the fifteenth week post fertilization, and this is the first tooth to emerge within the first 5 months after birth. All deciduous teeth are in occlusion by around 3 years of age. The first deciduous teeth to be shed are generally the central and lateral incisors around 7 years of age, when the permanent incisors emerge. The last deciduous tooth to be shed is generally the second molar in the tenth year. The first permanent tooth to show mineralization is the first molar, which occurs around the time of birth (sometimes earlier and sometimes later); it is also the first permanent tooth to emerge at around 6 years of age, and it will reach occlusion by the end of the seventh year. The last permanent tooth to emerge is the third molar; the variability of this occurrence makes it of restricted value for age prediction.

The time taken for a crown to form can be calculated from ground sections with considerable accuracy by counting the number of daily cross-striations from the neonatal line. For permanent teeth, the time taken for the crown to form can be calculated by counting the number of the enamel striae and multiplying by the individual's periodicity. Patterns of mineralization, emergence and shedding may be supplemented by analyses of tooth length, cementum apposition, secondary dentine formation, incremental enamel lines, attrition rates, root translucency and dentine transparency (Whittaker and MacDonald 1989). Chemical analysis of the stable isotopes of carbon, oxygen and nitrogen from teeth can provide important information concerning the nature of the diet of an individual, and is used in paleodiet analysis and in studies of pathophysiology and nutrition in archaeological and living human populations (Reitsema 2013).

DETERMINATION OF ETHNIC ORIGIN

The determination of ethnic or genetic origin is particularly difficult to achieve, although it is something that both physical and forensic anthropologists are frequently asked to attempt. The traditional view of ethnicity is that it is 'one of the major zoological subdivisions of mankind, regarded as having a common origin and exhibiting a relatively constant set of physical traits' (Bamshad and Olson 2003). Classifying groups on this basis is rather restrictive and, in our migrant modern world, somewhat artificial. It is still useful to be able to attempt to assign a 'most likely' genetic group, especially when dealing with unidentified forensic remains, but there are significant areas of overlap between the characteristics, and within any ethnic group there is often a full spectrum of representation. Yet we persist in classification on the basis of visual characteristics, and the area of the body that is most often analysed in this way is the skull.

Early anthropologists classified humans largely through geographical origins and recognized physical traits. The four traditional races of humans were: Caucasoid, Negroid, Mongoloid and Australoid. The Caucasoid race is geographically from Europe, North Africa, the Middle East, the Indian subcontinent and parts of Central Asia. Classically, the Caucasoid skull has a rounded to long shape (dolichocephalic) with a narrow nasal aperture, moderately developed supraorbital ridging, a prominent nasal spine, a steeple-shaped nasal root, little prognathism and a narrow interorbital distance. The forehead is steep, the chin is prominent, the palate is long and narrow, the cheek bones are not overly prominent, and there is a tendency to maxillary protrusion or mandibular retrusion. The Negroid race is geographically represented by Saharan, sub-Saharan and West African groups. The typical Negroid skull is also long with a wide nasal aperture, strong alveolar prognathism, low nasal root, guttering of the nasal aperture and a wide interorbital distance. The forehead is rounder, the palate is wider and the teeth are larger. The Mongoloid race is geographically represented by groups in East Asia, South East Asia, Central Asia, the Americas, Greenland, Inuit regions, Polynesia, South Asia and Eastern Europe. The typical Mongoloid skull is generally described as round with a nasal aperture of medium width, well-developed and high cheek bones, moderate prognathism, a tented nasal root, short nasal spine and shovel-shaped incisors. The palate is foreshortened, the forehead is vertical, the nasal bridge is low and there is a tendency for a forward rotation of the mandible. The Australoid race is geographically represented by Australian Aborigines, Maori, Pacific Islanders, Fijians and Papuans. The skull

of this rather heterogeneous group is generally represented by a broad nasal aperture, well-developed supraorbital ridging and glabella, and a wide palate with large teeth.

There is a current resurgence in research into ethnic determination that is largely centred on genetic markers, and consequently, skeletal indicators now play a significantly reduced role in this task (Gill and Rhine 2004).

FACIAL APPROXIMATION (RECONSTRUCTION)

Once the biological identity of an individual has been established (i.e. sex, age, stature and ethnic origin), an attempt to establish personal identity may be necessary, particularly in relation to a forensic investigation. Achieving a possible name for a victim is necessary before comparison of antemortem and postmortem data can be achieved and a positive identity established. One of the accepted ways to achieve this is to reconstruct the face from the skull, thereby producing a facial approximation that can be released to the community or the public at large in an attempt to identify the victim. Experts in this discipline utilize the many variations in the skull to reconstruct individual possible representations of the face from the material available. Biological variation will dictate the appropriate data used for tissue thickness that are applied to the skeletal scaffold.

There are fundamentally two approaches to facial reconstruction. The first approach is computer reconstruction – the skull is usually scanned by a laser three-dimensional scanner and an 'average' virtual face is wrapped around the skeletal scaffold. This approach is largely automated and requires limited training and expertise. It is rapid to achieve and relatively inexpensive, but relies on a large data set to ensure that the 'average' face utilized is appropriate. The second approach is modelled reconstruction – the skull is usually cast, and pegs are inserted into the cast at the appropriate tissue depth requirements. In the 'American' approach, a skin of clay is then moulded over the pegs to approximate the face. In the 'Manchester' method, each sequential muscle and soft tissue layer is built up around the pegs before a clay skin is moulded over the underlying structures. The modelling approach is clearly more dependent on experience, takes longer to achieve and is more costly.

There is great debate over which approach is the most accurate and, as yet, no agreement has been reached. Most practitioners state that the process achieves an approximation of one of the potential faces for the deceased and does not purport to reconstruct the actual face. The rationale is to produce an image that will jog the memory of the public and provide some possible names that will allow comparison of antemortem and postmortem identity data.

FACIAL SUPERIMPOSITION

Once a possible name has been derived, it may be necessary to compare the skull with photographs of the suspected individual. In these circumstances, an image of the skull is superimposed on to an image of the face of the missing person (Fig. 27.9). This relies on achieving a live capture image of the skull so that it can be rotated and manipulated into an identical position and to an identical size as the photograph. Features that do not change are lined up; a photograph that shows teeth is ideal because teeth can be lined up with the dentition on the skull. The image of the skull and photograph can then be faded in and out; if this is undertaken at speed, any discrepancies will show up on the image as distortion.



Bonus e-book images

Fig. 27.6 E–G, Normal three-dimensional calvarial anatomy. **E**, Frontal aspect. **F**, Lateral aspect. **G**, Basal aspect; there is a linear fracture in the right parietal bone.



Fig. 27.9 Facial superimposition: a live capture image of the skull has been manipulated into the identical position and size of the face in the photograph. (Courtesy of Professor Caroline Wilkinson and Professor Sue Black, Centre for Anatomy and Human Identification, University of Dundee.)

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